Lesson: Precalculus Review

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Core Precalculus Concepts

- Absolute Value
- Algebra Skills
- Domain/Range Mastery
- Trigonometry

Absolute Value

- $|a| = |-a|$
- $|ab| = |a||b|$
- $|x| < r \rightarrow -r < x < r$ if r is in real nums
- $|x| > r \rightarrow x > r \cup x < -r$ for r in real nums
- (make sure there isn't a domain restriction on x)

Absolute Value

$$\left|\left||x-2|-3\right|-1\right|=3$$

QUERY

WRITE AS PIECEWISE FUNCTION AND GRAPH:

$$f(x) = -|x+3| + |x-1|$$

Query

Find the parity [even, odd, or neither] of $f(x) = x^3 + \arcsin(\sin(x))$

QUERY

Solve $C = \{x \in \mathbb{R} \mid |4x+33| + 1 > 8\}$

Algebra Skills

- Know how to solve systems of equations with multiple different variables
  - Quick tricks!
  - Substitution
  - Elimination
  - Pattern-recognition

Practice Questions

1. $3x + 2y = 11$
   $3x + y = 7$

2. $3x + 2y = 12$
   $x + 2y = 8$

3. $3x - y = 10$
   $2x + y = 5$

4. $x - 3y = -17$
   $2x + 3y = 11$

Practice Questions - Answers

1

$$3x + 2y = 11$$
$$3x + y = 7$$

$x = 1, y = 4$

2

$$3x + 2y = 12$$
$$x + 2y = 8$$

$x = 2, y = 3$

3

$$3x - y = 10$$
$$2x + y = 5$$

$x = 3, y = -1$

4

$$x - 3y = -17$$
$$2x + 3y = 11$$

$x = -2, y = 5$

Domain and Range

Domain is the set of x-values, or inputs.

Range is the set of y-values, or outputs.

Let's crack some doubts!

Work with a partner to try the following practice problems and raise your hand to share.

QUERIES

4 JEE Main 2024 (Online) 8th April Morning Shift

Numerical +4 -1

If the range of $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta}, \theta \in \mathbb{R}$ is $[\alpha, \beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to ________.

$$f(x) = \arcsin \sqrt[3]{\ln(x^2 - x - 6)}$$

If domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right)+\cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ is $(\alpha, \beta) \cup (\gamma, \delta]$, then $18\left(\alpha^2+\beta^2+\gamma^2+\delta^2\right)$ is equal to ________.

Important Trig Identities & Properties To Know

1. $\sin\alpha = \frac{a}{c}; \quad \cos\alpha = \frac{b}{c}; \quad \operatorname{tg}\alpha = \frac{a}{b}; \quad \operatorname{ctg}\alpha = \frac{b}{a};$
   (a, b - legs, c - hypotenuse, α - angle opposing leg a).

2. $\operatorname{tg}\alpha = \dfrac{\sin\alpha}{\cos\alpha}; \quad \operatorname{ctg}\alpha = \dfrac{\cos\alpha}{\sin\alpha}.$

3. $\operatorname{tg}\alpha \operatorname{ctg}\alpha = 1.$

4. $\sin\left(\frac{\pi}{2} \pm \alpha\right) = \cos\alpha; \quad \sin(\pi \pm \alpha) = \mp\sin\alpha.$

5. $\cos\left(\frac{\pi}{2} \pm \alpha\right) = \mp\sin\alpha; \quad \cos(\pi \pm \alpha) = -\cos\alpha.$

6. $\operatorname{tg}\left(\frac{\pi}{2} \pm \alpha\right) = \mp\operatorname{ctg}\alpha; \quad \operatorname{ctg}\left(\frac{\pi}{2} \pm \alpha\right) = \mp\operatorname{tg}\alpha.$

7. $\sec\left(\frac{\pi}{2} \pm \alpha\right) = \mp\operatorname{cosec}\alpha; \quad \operatorname{cosec}\left(\frac{\pi}{2} \pm \alpha\right) = \sec\alpha.$

8. $\sin^2\alpha + \cos^2\alpha = 1.$

29. $\sin\alpha \pm \sin\beta = 2\sin\dfrac{\alpha \pm \beta}{2}\cos\dfrac{\alpha \mp \beta}{2}.$

30. $\cos\alpha + \cos\beta = 2\cos\dfrac{\alpha + \beta}{2}\cos\dfrac{\alpha - \beta}{2}.$

31. $\cos\alpha - \cos\beta = -2\sin\dfrac{\alpha + \beta}{2}\sin\dfrac{\alpha - \beta}{2}.$

32. $\operatorname{tg}\alpha \pm \operatorname{tg}\beta = \dfrac{\sin(\alpha \pm \beta)}{\cos\alpha\cos\beta}.$

33. $\operatorname{ctg}\alpha \pm \operatorname{ctg}\beta = \dfrac{\sin(\beta \pm \alpha)}{\sin\alpha\sin\beta}.$

34. $\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)].$

35. $\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)].$

36. $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha + \beta) + \cos(\alpha - \beta)].$

Product-to-sum, sum-to-product

Problem 3.25: Compute $\dfrac{\sin 13^\circ + \sin 47^\circ + \sin 73^\circ + \sin 107^\circ}{\cos 17^\circ}$. *(Source: ARML)*

3.4.4 Express

$$\frac{\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \sin 40^\circ + \sin 50^\circ + \sin 60^\circ + \sin 70^\circ + \sin 80^\circ}{\cos 5^\circ \cos 10^\circ \cos 20^\circ}$$

without using trigonometric functions. *(Source: HMMT)*

3.4.6★ Find the smallest positive integer solution to

$$\tan 19x^\circ = \frac{\cos 96^\circ + \sin 96^\circ}{\cos 96^\circ - \sin 96^\circ}.$$

(Source: AIME) **Hints:** 282, 24

More tough questions

Let a be a solution for c such that a > 0 and $3x^7 - 4c^2x^3 + cx^2 - c$ can be divided evenly by $x + 1$. Then arcsin(a) has a value of ______ radians? Express your answer as the number that corresponds to your choice.

1. $\frac{\pi}{6}$
2. $\frac{\pi}{4}$
3. $\frac{\pi}{3}$
4. $\frac{2\pi}{3}$
5. π
6. None of these

If the solution of the equation

$\log_{\cos x} \cot x + 4 \log_{\sin x} \tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha+\sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha + \beta$ is equal to :

The number of solutions of the equation

$$\log_{(x+1)}(2x^2 + 7x + 5) + \log_{(2x+5)}(x + 1)^2 - 4 = 0, x > 0, \text{ is :}$$

Lesson: Limits Review

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Core Limits Concepts

- What are Limits?
- Formal Definition Of Limits
- Basic Limits Laws
- Solving Limits
- squeeze theorem
- Continuity
- Intermediate Value Theorem

What is a limit?

A limit is…

DEFINITION OF LIMIT

We are going to learn the precise definition of what is meant by the statements

$$\lim_{x \to a} f(x) = L$$

and

$$\lim_{x \to a} f(x) = +\infty \qquad \lim_{x \to a} f(x) = -\infty .$$

We must make precise our intuitive notion that $f(x)$ gets arbitrarily close to L as x gets close to a .

FORMAL DEFINITION OF LIMITS

Epsilon-delta

$$\lim_{x \to 3} (4x - 1) = 11$$

$$\forall \varepsilon, \exists \delta : 0 < |x - a| < \delta$$

Basic Limit Laws

Limit Laws

If $\lim_{x \to a} f(x) = M$ and $\lim_{x \to a} g(x) = N$

Sum	$\lim_{x \to a} \left[f(x) + g(x) \right] = M + N$	
Difference	$\lim_{x \to a} \left[f(x) - g(x) \right] = M - N$	
Constant	$\lim_{x \to a} \left[k \bullet f(x) \right] = k \bullet M$	
Product	$\lim_{x \to a} \left[f(x) g(x) \right] = M \bullet N$	
Quotient	$\lim_{x \to a} \left[\dfrac{f(x)}{g(x)} \right] = \dfrac{M}{N}$	$N \neq 0$
Power	$\lim_{x \to a} \left[f(x) \right]^n = M^n$	n is a positive integer
Root	$\lim_{x \to a} \left[\sqrt[n]{f(x)} \right] = \sqrt[n]{M}$	n is a positive integer

How To Solve For limits

1. JUST PLUG IT IN!!!
2. If that gives you a indeterminate form, then try to manipulate the function.
    a. Cancel the factor leading the denominator to being zero.
    b. Apply trig properties
    c. Noticing patterns

Algebraic Limit-Solving

$$\lim_{x\to2}\frac{8-x^3}{x^2-4}$$

$$\lim_{t\to -1}\frac{2-\sqrt{t^2+3}}{t+1}$$

$$\lim_{x\to -5}\frac{x+7}{x^2+3x-10}$$

Pattern Recognition

$$lim\left(x \to \frac{\pi}{2}\right)\left[(1 + \cot x)^{\tan x}\right]$$

Continuity

Continuity

A function f **is continuous at a number a** if:

1. $f(a)$ is defined

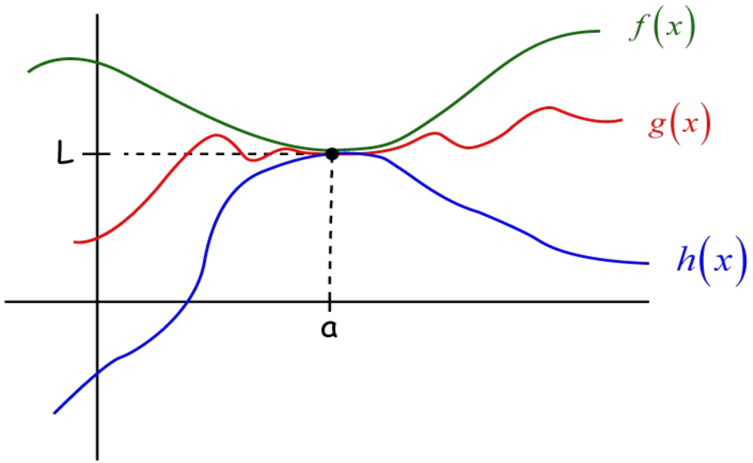
2. $\lim_{x \to a} f(x)$ exists

3. $\lim\limits_{x \to a} f(x) = f(a)$

Squeeze Theorem:

If $h(x) \le g(x) \le f(x)$ when x is near a, except possibly at a,

and $\lim\limits_{x \to a} f(x) = \lim\limits_{x \to a} h(x) = L$, then $\lim\limits_{x \to a} g(x) = L$.



Squeeze Theorem:

$$\lim_{x \to -\infty} \frac{4x^2 - \sin(5x)}{x^2 - 7}$$

IVT

Intermediate Value Theorem

If f is continuous on $[a,b]$ and k is any number between f(a) and f(b), inclusive, then there is at least one number c in the interval [a,b] such that f(c) = k.

More Practice

Use the Intermediate Value Theorem to prove that the equation $x^3 = 2x^2 + 3x - 3$ has three solutions.

Evaluate $\displaystyle\lim_{x \to 0} \frac{\sqrt{1+x} - 1}{x}$

Evaluate $\displaystyle\lim_{x \to 0} \frac{x}{\sqrt{a+x} - \sqrt{a-x}}$

R4C3-2-F!N!SH

a) $\displaystyle\lim_{x\to\infty}\frac{x^2-1}{2x^2+1}$

b) $\displaystyle\lim_{x\to-\infty}\frac{x^3+x^2-4}{2x^3+x+11}$

c) $\displaystyle\lim_{x\to\infty}\frac{3x^2+2x-1}{x^3-x+2}$

d) $\displaystyle\lim_{x\to\infty}\left(\frac{x^3}{x^2+2}-x\right)$

e) $\displaystyle\lim_{x\to\infty}\frac{x^2+3x-4}{3x^2-2x+5}$

f) $\displaystyle\lim_{x\to\infty}\frac{x(x-1)(x-2)}{x^2+6x-9}$

g) $\displaystyle\lim_{x\to\infty}\frac{\sqrt{x^2+9}}{x+3}$

h) $\displaystyle\lim_{x\to\infty}\left(\frac{x^2+x-1}{2x^2-x+1}\right)^3$

i) $\displaystyle\lim_{x\to\infty}\frac{x^2+2x+1}{5x}$

j) $\displaystyle\lim_{x\to-\infty}\frac{x^3+x^4-1}{2x^5+x-x^2}$

k) $\displaystyle\lim_{x\to\infty}\frac{(\sqrt{x^2+1}+x)^2}{\sqrt[3]{x^6+1}}$

l) $\displaystyle\lim_{x\to-\infty}\frac{x^6+7x^4-40}{1-x-5x^7}$

m) $\displaystyle\lim_{x\to\infty}\frac{(x+1)(x-2)}{3x^2+6x-5}$

n) $\displaystyle\lim_{x\to\infty}\frac{\sqrt{x^2+1}}{x}$

o) $\displaystyle\lim_{x\to\infty}\left(\frac{3x^2+2x+1}{x^2-3x+2}\right)^4$

p) $\displaystyle\lim_{x\to-\infty}\frac{5x^3-x^2+x}{1-x-3x^2}$

q) $\displaystyle\lim_{x\to\infty}\frac{1+x-3x^3}{1+x^2+3x^3}$

r) $\displaystyle\lim_{x\to-\infty}\left(\frac{x^3-8}{x^4+16}\right)^{10}$

s) $\displaystyle\lim_{x\to\infty}\frac{(x+3)(x+4)(x+5)}{x^4+x-11}$

t) $\displaystyle\lim_{x\to-\infty}\frac{8x-2x^5+x^6}{11x+5x^3+3x^5}$

u) $\displaystyle\lim_{x\to\infty}\left(\frac{x^3}{2x^2-1}-\frac{x^2}{2x+1}\right)$

v) $\displaystyle\lim_{x\to\infty}\left(x^2-\frac{x^4-1}{x^2-2}\right)$

w) $\displaystyle\lim_{x\to\infty}\frac{(x-1)^{100}(6x+1)^{200}}{(3x+5)^{300}}$

x) $\displaystyle\lim_{x\to\infty}\frac{\sqrt[4]{x^5}+\sqrt[5]{x^3}+\sqrt[6]{x^8}}{\sqrt[3]{x^4+2}}$

y) $\displaystyle\lim_{x\to-\infty}\frac{x^2(2x+1)(3x-2)}{2x^2(5x-8)(x+6)}$

z) $\displaystyle\lim_{x\to\infty}\left(\frac{2x^8+8x^6+6x^4}{4x^8-x^6+12x^4}\right)^5$

Z) $\displaystyle\lim_{x\to-\infty}\frac{(2x-3)^{20}(3x+2)^{30}}{(2x+1)^{50}}$

TRY OUT THESE FUN INFINITE LIMITS AND NOTICE SOME KEY PATTERNS.

TURN AND TALK.